Spatiotemporal Analysis of Venezuela's Nighttime Light During the Socioeconomic Crisis

Lin Zhang, Xi Li, and Fengrui Chen

***Abstract*— The lack of official statistics makes it difficult to assess Venezuela's economic situation during the socioeconomic crisis. In this article, we used Visible Infrared Imaging Radiometer Suite (VIIRS) nighttime light images to evaluate Venezuela's crisis. The Hodrick–Prescott filter was used to decompose the sum of urban light (SUL) into two components: the sum of urban light trend (SULT) and the SUL cycle. Then, we proposed an index of nighttime light change ratio to estimate Venezuelan nighttime light dynamics. We found that Venezuela has lost 30.37% of its SULT from April 2012 to December 2018. The regression analysis shows that Venezuelan SULT had a strong relationship to a number of socioeconomic indicators: the SULT was positively correlated to crude oil production with R^2 of 0.9159, negatively correlated to dollar exchange rate with R^2 of 0.9516, and negatively correlated to the number of asylum seekers with R^2 of 0.8384. We also found that among the three states with a largest nighttime light decrease, the economy of two states was dominated by agriculture and that of one state was dominated by the oil industry. In the pixel analysis, compared with the urban cores, the suburbs of urban cores of 12 main cities had a higher percentage of SULT increased areas. Around the Venezuela–Colombia border, the SULT decreased in the Venezuelan side but increased in the Colombian side. Our analysis suggests that nighttime light imagery can help to assess Venezuela's situation during the crisis.**

***Index Terms*—Nighttime light, oil production, socioeconomic crisis, spatial analysis, Venezuela.**

I. Introduction

A SOCIOECONOMIC and political crisis that began in Venezuela in 2010 has been marked by hyperinflation, disease, food shortage, climbing murder rates, increased death rates, and massive emigration from the country. This has been the worst socioeconomic crisis in Venezuelan history and among the worst crises experienced in the Americas. By 2017, this crisis had affected the life of the Venezuelans on all levels [1]. Almost three-fourths of the population had lost over 8 kg in weight [2], and more than half Venezuelan did not have enough income to meet their basic food needs. In 2018, 94% of Venezuelans lived in poverty, and over 3.4 million people (10% of Venezuelans) have left their country [3]. And about 5000 people left Venezuela every day in search of protection or a better life; the vast majority of them (2.7 million) were hosted in countries of the Caribbean and Latin America, especially Colombia [4]. Venezuelan climbing murder rates also led the world, with 56.3 per 100 000 people killed in 2016, making it the third most violent country in the world [5]. The United Nations estimated that one-quarter of Venezuelans needed humanitarian assistance in 2019 [6].

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Venezuelan government stopped publishing official inflation figures in early 2018. Given conditions on the ground, although the World Bank predicts the Venezuelan annual inflation index since 2013, it is hard to establish clear empirical indicators of how things have been going in Venezuela. It is a prompting urgency to evaluate the socioeconomic condition in Venezuela from other data sources. Remote sensing as an objective, low cost, and large coverage tool has been widely used to assess conflicts of global areas, such as Kuwait [7], [8], Bosnia [9], Chechnya [10], and Congo [11]. As a unique remote sensing data source, nighttime light images can reflect human activities, because human settlements, where economic activity takes place, emit light at night. Nighttime light images can be used in socioeconomic studies, including carbon emission estimation [12], poverty and inequality assessment [13], [14], GDP estimation [15], [16], urbanization mapping [17]–[19], and play an important role in conflict evaluation, such as the cases in Iraq [20], Syria [21], [22], and Yemen [23]. Some experts also used nighttime light images to evaluate light pollution [24] and its impact on human health [25].

In this article, we addressed three main issues. First, we proposed two indicators of nighttime light: the sum of urban light trend (SULT) and nighttime light change ratio (NLCR). Second, we explored the relationship between the SULT and three socioeconomic indicators, namely crude oil production, exchange rate and the number of asylum seekers. Third, we used NLCR to evaluate Venezuelan nighttime light change patterns from a spatiotemporal perspective and its response to the crisis.

II. Methodology and Method

Venezuela is located in the north of South America and has 23 states (Estados), one Capital District (Distrito Capital, D.C.), and the Federal Dependencies (Dependencias Federales

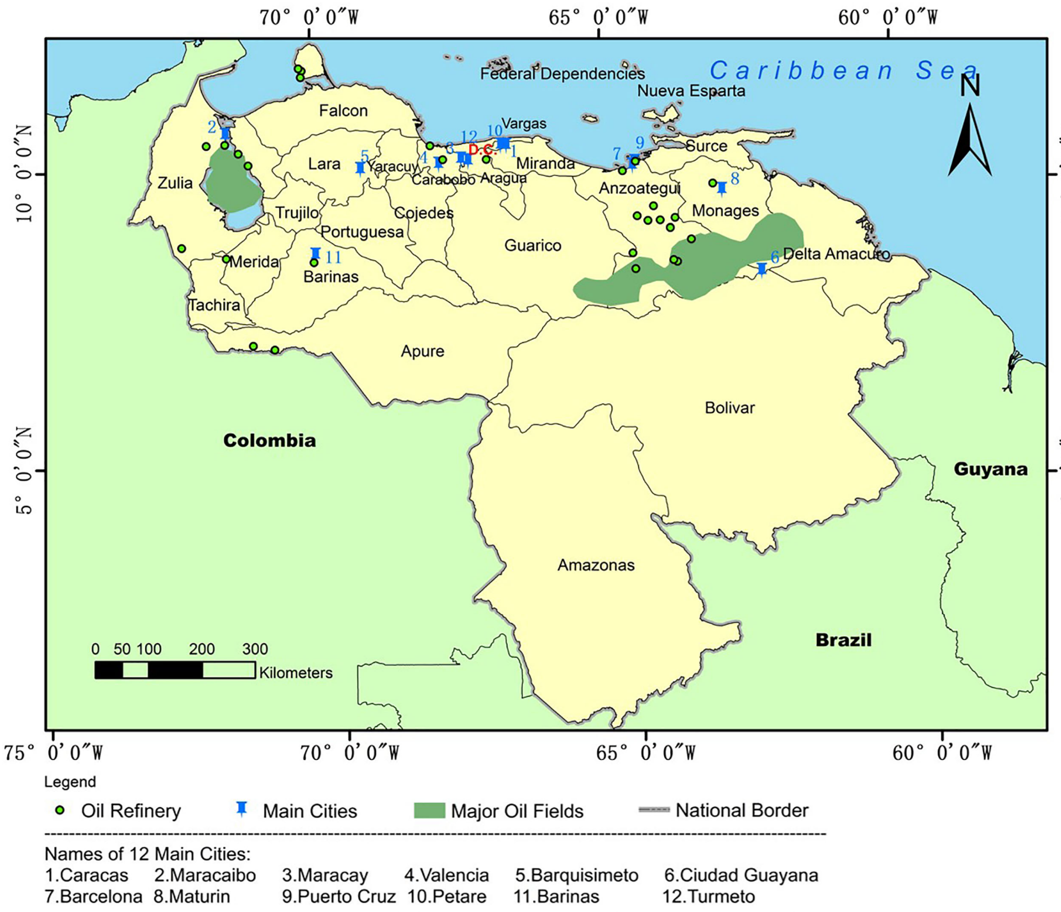

Fig. 1. Location of Venezuela.

de Ultramar), including Venezuelan islands (Fig. 1). Venezuela borders the Caribbean Sea to the North, Colombia to the West, Brazil to the South, and Guyana to the East. The territory extends between the equator and 13° North, and 59° to 74° West.

A. *Materials of Study*

The Suomi National Polar-Orbiting Partnership Visible Infrared Imaging Radiometer Suite (NPP–VIIRS) has been widely used in nighttime light remote sensing. The NPP–VIIRS recorded its first image in November 2011 with good radiometric detection capacity, without radiometric saturation in the urban core areas [26]. We used VIIRS nighttime light monthly radiance composite product as our data source, as an example shown in Fig. 2. The 81 nighttime light images cover the period from April 2012 to December 2018. Because Venezuela does not provide reliable official statistics, we used socioeconomic statistical data and auxiliary data sets from independent organizations, as described in Table I.

Because Venezuela is an oil-rich country, gas flares related to oil production can be observed in the VIIRS nighttime light images. Therefore, the values of pixels with gas flares are much higher than those of urban pixels. We used an urban extent map

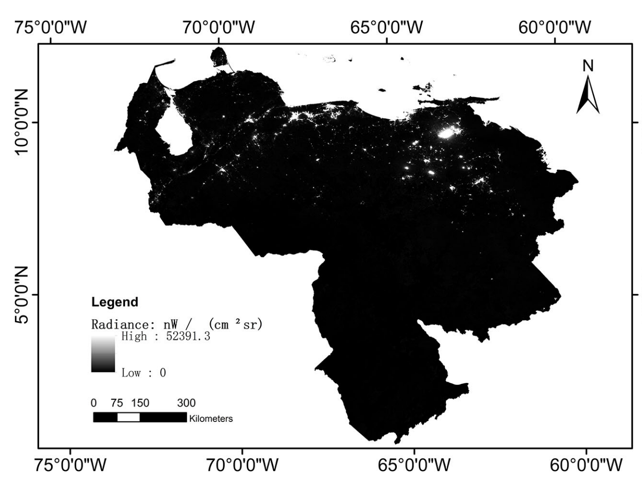

Fig. 2. VIIRS nighttime light image of Venezuela in April 2012.

derived from the Global Urban Footprint (GUF) [36] product to mask out the nonurban light in Venezuela (Fig. 3).

Since clouds block the satellite from receiving nighttime light during the rainy season, there were some no-data pixels in the images, so it is necessary to complement no-data pixels

TABLE I
DATA SETS USED IN THIS STUDY

Data	Source
VIIRS monthly average radiance composite product	Version 1 VIIRS Day/Night Band Nighttime Lights [27]
Administrative boundary map	GADM maps and data [28]
Global urban footprint	DLR Earth Observation Center [29]
Crude oil production	U.S. Energy Information Administration [30]
Exchange rate	Dollar Today [31]
The number of asylum seekers	United Nations High Commissioner for Refugees [32]
Names and locations of oil refineries	Google Maps [33]
Major oil fields	The Energy Consulting Group [34]
Venezuelan population	World Population Review [35]

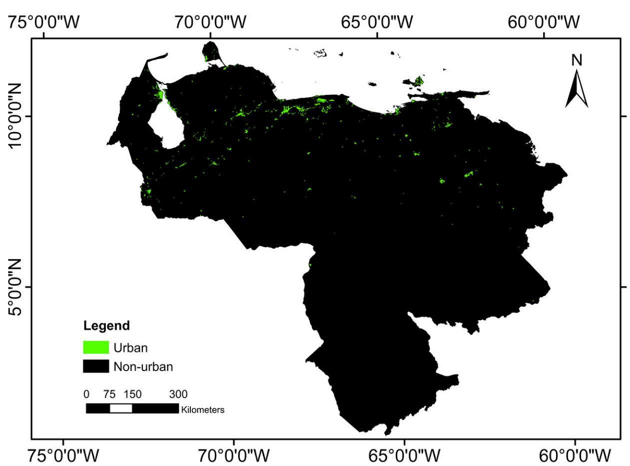

Fig. 3. Global Urban Footprint (GUF) map for Venezuela.

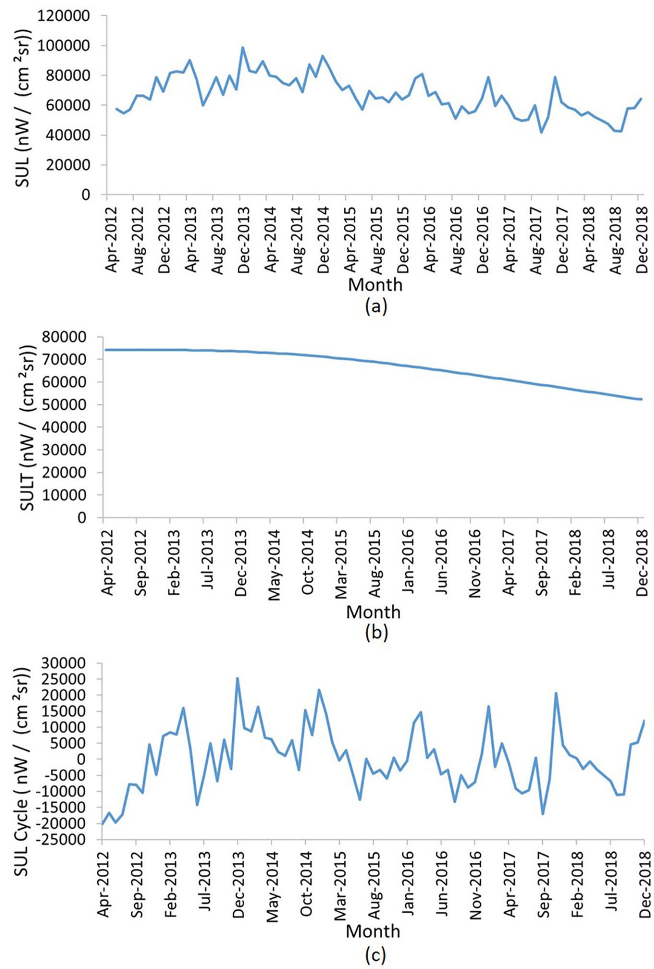

Fig. 4. Decomposition of the sum of urban light (SUL) of Anzoategui state. (a) Sum of urban light (SUL). (b) Sum of urban light trend (SULT). (c) Sum of urban light cycle (SUL cycle).

for further quantitative analysis. We used linear regression to interpolate the no-data pixels in the time-series images. The original multitemporal VIIRS image contains 5 616 137 pixels and 6391 urban pixels. In the 6391 urban pixels, 1371 urban pixels (21.452%) have missing value. In addition, 40 urban pixels (0.626%) have missing data for two consecutive months, and no pixel has missing data for three consecutive months or more. After linear interpolation, the no-data urban pixels are interpolated.

B. Indices of Urban Nighttime Light

To get the time series of urban nighttime light, the sum of urban light (SUL) is used to calculate the urban nighttime light with the following equation:

$$\mathrm{SUL} = \sum_{i=1}^{n} r_i \tag{1}$$

where r_i denotes the radiance value of the ith urban pixel inside a region in the image.

However, Venezuelan nighttime light showed strong seasonal change patterns. It was reported that monthly nighttime light data are impacted by seasonal change in land surface reflectance, plant phenology, and atmospheric components [37]. To remove the seasonal change patterns, we used the Hodrick–Prescott filter (HP-filter) [38], [39] to decompose the nighttime light into two components: SULT and SUL cycle. The main idea of HP-filter is to obtain a smoothed curve to represent a time series (SULT) that is more sensitive to a long-term trend than to short-term fluctuation (SUL cycle). Fig. 4 shows the decomposition result of the SUL of the Anzoategui state as an example.

For each region, the SULT was calculated for all months and was used to derive the NLCR as

$$\mathrm{NLCR} = \frac{\mathrm{SULT}_{\mathrm{end}} - \mathrm{SULT}_{\mathrm{start}}}{\mathrm{SULT}_{\mathrm{start}}} \tag{2}$$

where $\mathrm{SULT}_{\mathrm{start}}$ and $\mathrm{SULT}_{\mathrm{end}}$ denote the SULT for the beginning month (e.g. April 2012) and the last month (e.g. December 2018), respectively. We used a Log–Log model [40], [41] from econometrics to develop a nonlinear regression between the SULT and socioeconomic indicators. The Log–Log model is

$$lnY = b_0 + b_1 lnX \tag{3}$$

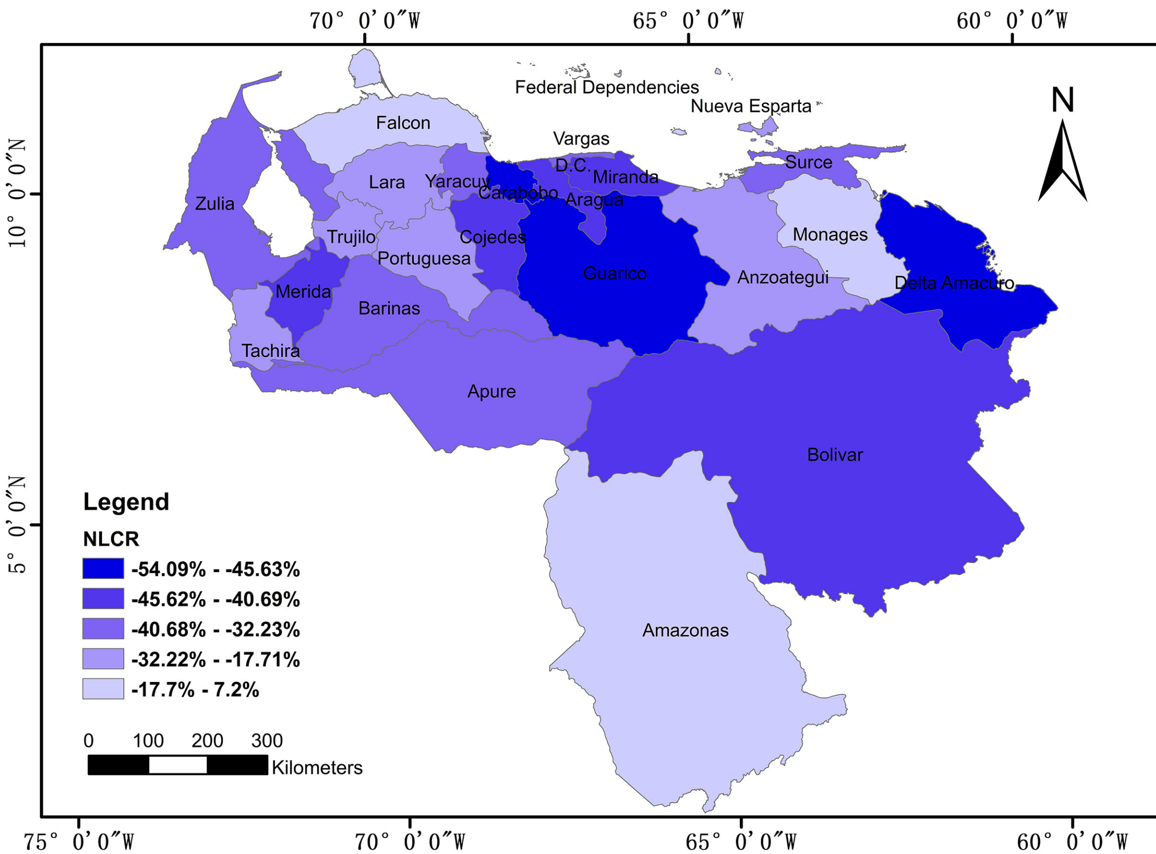

Fig. 5. Nighttime light change ratio (NLCR), derived from sum of urban light trend of Venezuela's 24 administrative areas.

and we differentiate it to

$$\frac{\delta Y}{Y} = b_1 \frac{\delta X}{X}. \quad (4)$$

The term on the right-hand side is the change rate in X, and the term on the left-hand side is the change rate in Y. This equation means when X increases by 1 percentage, Y increases by b_1 percentage.

III. Results and Analysis

A. Nighttime Light Change Patterns at National and State Levels

We found that from April 2012 to December 2018, Venezuelan urban area lost 30.37% of its SULT. To explore the pattern of nighttime light at the state level, we analyzed the nighttime light dynamics in the administrative areas (Figs. 5 and 6). Since Federal Dependencies comprises a large number of islands and has no urban region, we did not analyze it. The NLCR of Venezuelan 24 administrative areas is listed in Table II. Overall, the SULT decreased in all 24 administrative areas, and 19 administrative areas lost over a third of their SULT. In the three states with largest decrease of nighttime light, Delta Amacuro and Guarico were the main agricultural states of Venezuela. Agriculture is the main industry of these two states [42], [43]. Delta Amacuro had lost 54.09% of its SULT, and Guarico had lost 46.16% of its SULT. During the Venezuelan crisis, an exodus of people led to a sharp decline in the rural population, resulting in a decline of agricultural output. It further affected the Delta Amacuro and Guarico's economy, then probably led to the nighttime light decrease. In Carabobo, it had lost 48.13% of its SULT, the oil industry was the main economic activity [44], and we infer that the reduction of crude oil production had heavily influenced Carabobo's economy, which also probably led to SULT decrease. These findings coincided with the Venezuela

TABLE II
NLCR of 24 Administrative Areas

State	NLCR	State	NLCR
Delta Amacuro	-54.09%	Zulia	-34.79%
Carabobo	-48.13%	Vargas	-33.47%
Guarico	-46.16%	Yaracuy	-33.12%
Distrito Federal	-45.11%	Tachira	-32.19%
Miranda	-44.91%	Portuguesa	-31.77%
Aragua	-42.34%	Lara	-30.88%
Bolivar	-42.11%	Nueva Esparta	-30.27%
Cojedes	-40.75%	Anzoategui	-29.52%
Merida	-39.00%	Trujillo	-28.98%
Barinas	-37.86%	Amazonas	-8.93%
Apure	-35.53%	Monagas	-2.45%
Surce	-35.07%	Falcon	-2.36%

All 24 administrative areas passed the Mann–Kendall trend test: 99% down.

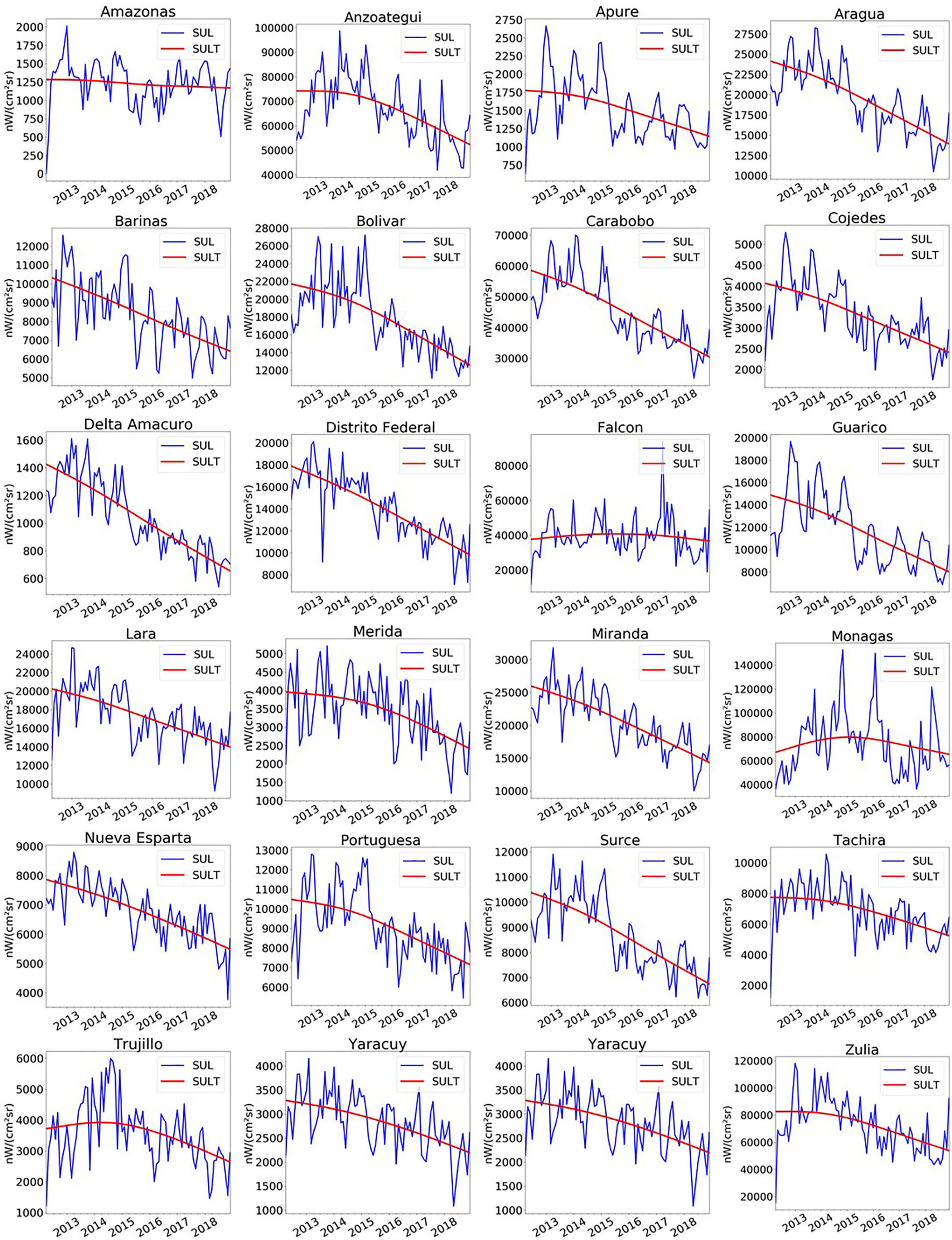

Fig. 6. Sum of urban light (SUL) and sum of urban light trend (SULT) of Venezuela's 24 administrative areas.

TABLE III
ATTRIBUTES OF 30 REFINERIES AND THEIR NLCR (PDVSA IS THE VENEZUELA STATE-OWNED COMPANY, NLCR DENOTES NIGHTTIME LIGHT CHANGE RATIO)

No.	Name	NLCR	No.	Name	NLCR
1	Unknown[1]	-19.93%	16	Procesadora Oropal[1]	-50.18%
2	Unknown[1]	-93.67%	17	Baripetrol Filial[1]	-12.33%
3	Unknown[1]	-80.98%	18	Patio Tanques Punta Gorda[1]	-36.97%
4	Unknown[1]	-87.06%	19	Edificio El Barco[1]	-40.15%
5	Unknown[1]	-75.55%	20	COG PDVSA[1]	-41.76%
6	Unknown[1]	-78.99%	21	Petroboscan[1]	-27.42%
7	Unknown[1]	-80.07%	22	VASSA GUACARA[1]	-45.86%
8	Unknown[1]	-61.00%	23	El Palito[1]	-19.04%
9	Unknown[1]	-97.58%	24	Puerto La Cruz[1]	-50.86%
10	Unknown[1]	-95.95%	25	Bajo Grande[2]	45.98%
11	Unknown[1]	-59.23%	26	San Roque[2]	77.91%
12	Unknown[1]	-91.03%	27	Amuay[2]	238.16%
13	Refinery Victoria[1]	-40.81%	28	Cardon[2]	47.79%
14	Mejorader Petropiar[1]	-92.32%	29	Planta de Distribution Gasolina[2]	93.40%
15	Bare Operating Center[1]	-57.52%	30	Planta de Distribution[2]	3.53%

[1]The Mann–Kendall trend test of nighttime light change: Down ($P < 0.01$); [2]The Mann–Kendall trend test: Up ($P < 0.01$).

TABLE IV
MAIN PRODUCTS OF SIX REFINERIES OF PDVSA COMPANY

Name	NLCR	Main production
El Palito[1]	-19.04%	Medium grade crude oil
Puerto La Cruz[1]	-50.86%	Light and heavy grade crude oil
Bajo Grande[2]	45.98%	Asphalt
San Roque[2]	77.91%	Paraffin
Amuay[2]	238.16%	Asphalt
Cardon[2]	47.79%	Asphalt

[1]The Mann–Kendall trend test of nighttime light change: Down ($P < 0.01$); [2]The Mann–Kendall trend test: Up ($P < 0.01$).

TABLE V
POPULATION AND NLCR OF 12 VENEZUELAN MAIN CITIES

City	Population (2019)	NLCR
Caracas	3,000,000	-46.29%
Maracaibo	2,225,000	-43.32%
Maracay	1,754,256	-41.95%
Valencia	1,385,083	-52.09%
Barquisimeto	809,490	-32.38%
Ciudad Guayana	746,535	-48.81%
Barcelona	424,795	-39.97%
Maturin	410,972	-46.40%
Puerto Cruz	370,000	-43.12%
Petare	364,684	-39.54%
Barinas	353,442	-39.54%
Turmero	344,700	-39.10%

All 12 main cities passed the Mann–Kendall trend test: Down ($P < 0.01$).

crisis that led to the Venezuela economic recession, which was reflected in the nighttime light loss in those areas. However, some states with concentrated oil industry did not show a large nighttime light decrease, and two examples are Monages and Falcon (Fig. 6).

TABLE VI
PIXEL ANALYSIS OF THE 60 KM BUFFER REGION AROUND THE VENEZUELA–COLOMBIA BORDER; UNIT: NUMBER OF PIXELS

Border Area	Lit Pixels	NLCR Increase	NLCR Decrease	Increase Ratio	Decrease Ratio
Venezuela	458	90[1]	343[2]	19.65%	74.89%
Colombia	407	252[1]	118[2]	61.92%	28.99%

[1]The Mann–Kendall trend test: Up ($P < 0.05$); [2]The Mann–Kendall trend test: Down ($P < 0.05$).

B. Oil Production and Venezuelan Economy

Venezuela is an oil-exporting country and highly depends on the import of consumer goods. Oil exports constituted about 95% of its total exports in 2014 [45]. Oil receipts accrue to the state-owned company, Petroleos de Venezuela (PDVSA), which transfers some of its earnings to the national budget and engages in quasi-fiscal activities focusing on social and developmental objectives. Oil export earnings comprised the main source of foreign exchange, which is used to import foods, consumer goods, and intermediate inputs for oil production. In 2014, when the international oil price fell, Venezuela faced a deficit in its foreign exchange earnings, which resulted in a cut of over 70% in imports [46].

Using (3), we found a positive and significant ($R^2 = 0.9159$, $p < 0.01$) relationship between SULT and Venezuela crude oil production (in the unit of BBL/D/1K; one thousand of barrels per day), as shown in Fig. 7(a). This analysis shows that SULT increased by 0.4991% when oil production increased by 1%. During the hyperinflation period, Venezuelan currency (Bolivar) devalued fast. Fig. 7(b) shows when the SULT decreased by 0.0304%, the exchange rate increased by 1%, and the R^2 of this nonlinear regression reached 0.9516.

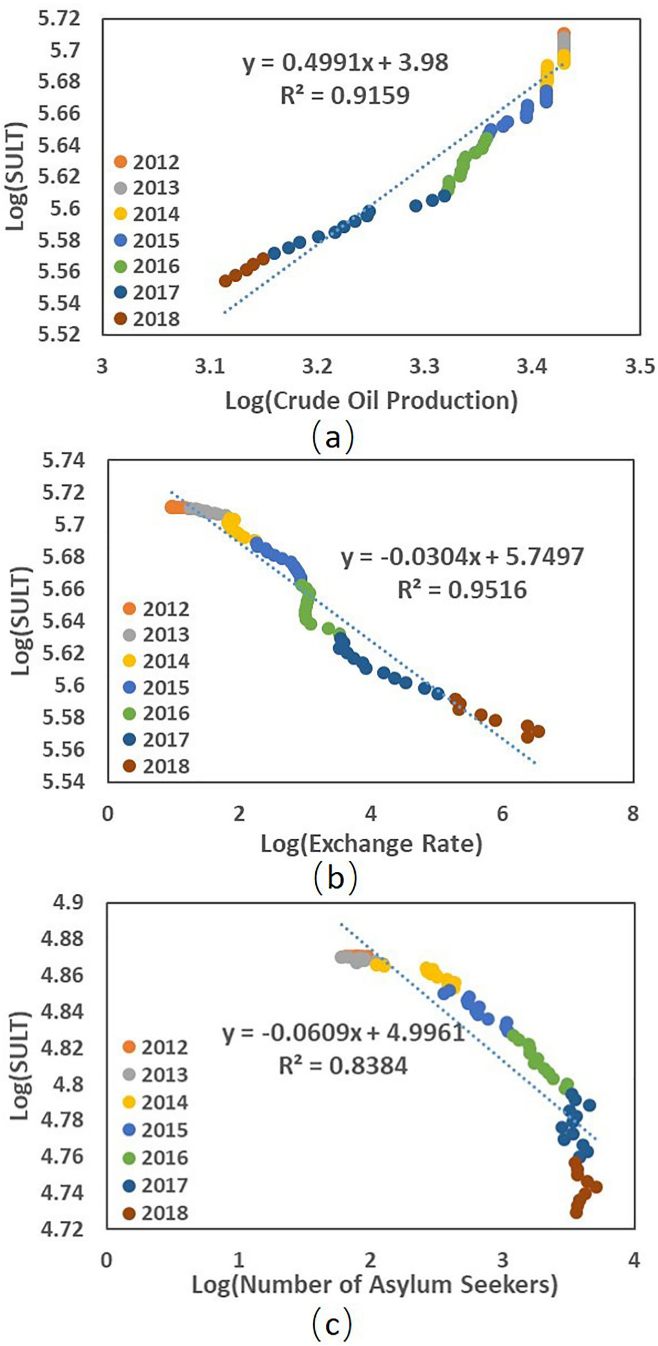

Fig. 7. Log–Log regression of sum of urban light trend (SULT) versus socioeconomic indicators for Venezuela. (a) SULT versus crude oil production. (b) SULT versus exchange rate. (c) SULT versus number of asylum seekers.

During the crisis, many Venezuelans became asylum seekers. We obtained the monthly number of Venezuelan asylum seekers from UNHCR. From Fig. 7(c), we found that the relationship between the Venezuelan urban SULT and the number of asylum seekers was negative and significant based on the Log–Log regression ($R^2 = 0.8384, p < 0.01$). As the number of Venezuelan asylum seekers increased, the Venezuelan urban nighttime light decreased. The results from Fig. 7 suggest that nighttime lights

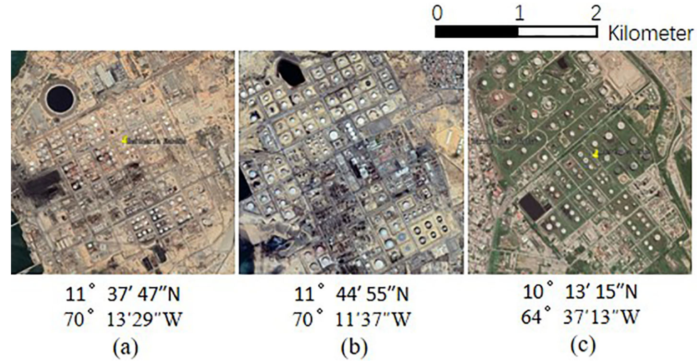

Fig. 8. Examples of the Venezuelan oil refinery in Google Earth. (a) Cardon. (b) Amuay. (c) Puerto La Cruz.

are able to serve as a proxy to estimate Venezuela's socioeconomic situation.

Using the nighttime light images combined with Google Earth, we geo-located 30 oil refineries in Venezuela, with examples shown in Fig. 8. The nighttime light image was used to find very bright regions that were potentially affected by gas flares. For visual interpretation, we used Google Earth to verify these regions including refineries. In Venezuela, refineries often distribute around oil wells. In the rest of this article, the nighttime light of refineries denotes the nighttime light in the refinery areas, including oil wells.

In Table III, we calculated the NLCR of the 30 refineries and we marked the ownership of these refineries in Fig. 9. We found that the refineries which had nighttime light increased were all associated with the PDVSA Company. Based on Table III, we are interested in the impact of the oil products of the refinery on nighttime light dynamics. In Table IV, we found only six refineries that were run by PDVSA Company had public-accessible product information, all derived from the official website of PDVSA[1].

These six refineries were destined for different oil products. From Table IV, four refineries showed nighttime light increase, and their main products are either asphalt or paraffin. Due to limited access to credit of PDVSA from U.S. sanctions that have disrupted its oil purchases and bank guarantees for oil intermediate inputs, we infer that PDVSA used the currently operating refineries mainly for the production of relatively simple oil products, such as asphalt and paraffin [47]. However, the two refineries (e.g., El Palito and Puerto La Cruz) with nighttime light decreased were mainly used to produce crude oil. Therefore, this finding confirms that crude oil production declined as a consequence of the U.S. sanction on the state-owned oil company (PDVSA).

C. Nighttime Light of Urban Cores and Suburbs

Venezuela has experienced several dark moments when most of the states stopped receiving electricity, internet service, and

[1] [Online]. Available: http://www.pdvsa.com

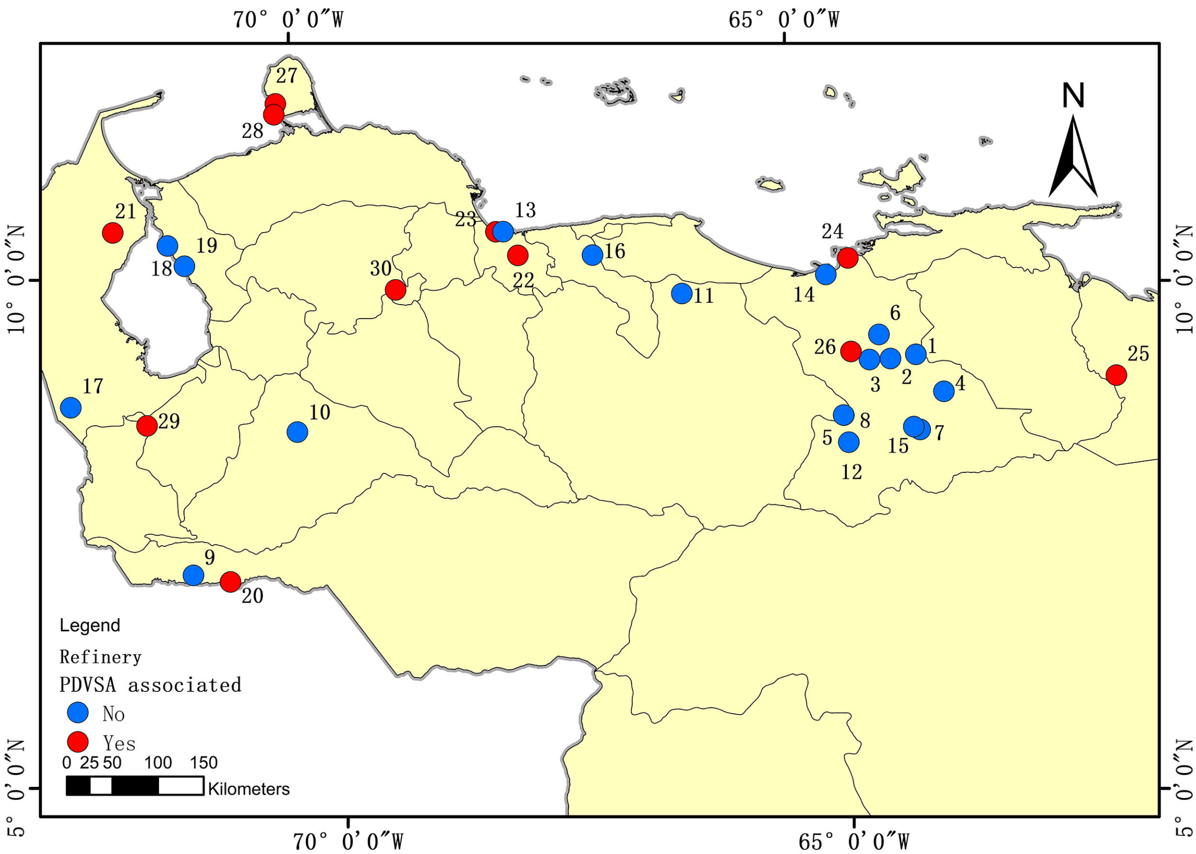

Fig. 9. Location of the 30 refineries.

TABLE VII
NLCR of 18 Venezuela–Colombia Border Cities

Country	Name	Location	Distance to Border (km)	Mann-Kendall trend test	NLCR
Colombia	Arauca	7°5′10″N 70°45′25″W	0.00	Up	58.17%
	Cucuta	7°54′33″N 72°30′23″W	0.00	Up	9.00%
	Maicao	11°22′54″N 72°14′43″W	8.77	Up	14.90%
	Puerto Carreno	6°11′3″N 67°29′11″W	0.00	Up	7.30%
	Saravena	6°57′8″N 71°52′33″W	4.42	Up	10.37%
Venezuela	Aguas Calientes	7°55′6″N 72°26′25″W	0.00	Down	-5.27%
	El Amparo	7°6′2″N 70°45′8″W	0.00	Down	-2.00%
	Guasdualito	7°13′52″N 70°43′49″W	12.57	Down	-37.96%
	Lobatera	7°57′1″N 72°14′27″W	12.28	Down	-7.33%
	Paraguaipoa	11°20′57″N 71°58′15″W	13.03	Down	-25.89%
	Rubio	7°41′38″N 72°21′56″W	9.80	Down	-57.24%
	San Cristobal	7°46′48″N 71°12′54″W	13.32	Down	-34.63%
	San Juan	8°2′31″N 72°15′1″W	7.88	Down	-62.68%
	Santa Ana	7°38′11″N 72°16′54″W	18.40	Down	-17.32%
	La Frita	8°13′25″N 72°15′0″W	10.14	No Tendency	6.33%

All passed the Mann–Kendall trend test: $P < 0.01$.

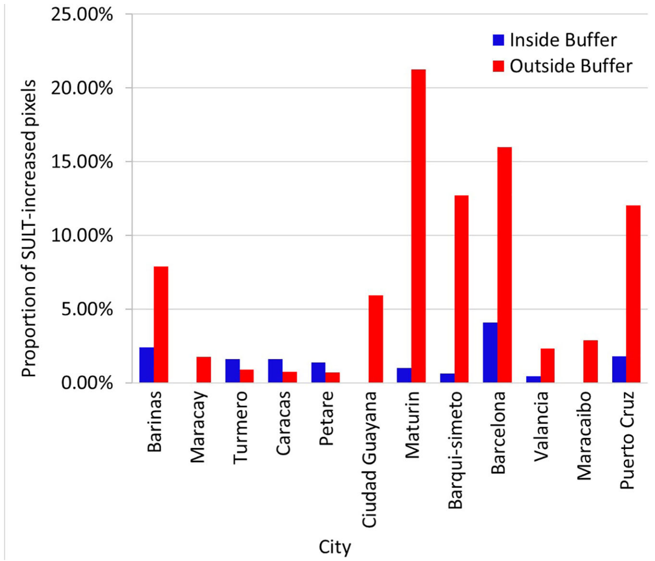

Fig. 10. Proportion of SULT-increased pixels of urban core and suburb of 12 main cities in Venezuela. (The inside buffer covered the pixels with distance to the urban core less than 10 km; the outside buffer covered the pixels with distance of 10–60 km to the urban core.)

cell phone signals. These events halted work, school activities as well as commercial activities, and both the metro and the railroad stopped working. The failures and collapse of basic services, such as electricity gas and water, are the reasons why Venezuelans sought refuge outside the urban area [48].

To find out the nighttime light change pattern at the city level, we selected Venezuelan 12 main cities to analyze their nighttime light, as shown in Table V. The total population of these 12 cities accounted for 42.82% of the Venezuelan population in 2019. Each city had lost over 30% of its SULT.

To find out the differences in the nighttime light change pattern between the urban core and suburb, we conducted the pixel analysis to calculate the NLCR of each pixel at a 750-m resolution. As the average urban radius of these 12 cities is less than 10 km, we used two different buffers to analyze the NLCR of the pixels in these cities. The inside buffer covered the pixels with distance to the urban core less than 10 km; the outside buffer covered the pixels with the distance between 10 and 60 km to the urban core (Fig. 10).

In 9 cities out of the 12 cities, compared with the urban core (inside buffer), the suburb (outside buffer) had a higher percentage of pixels with increased SULT. This difference in the nighttime light dynamics between urban core and suburb implies that residents from the urban cores moved outward to seek safer living conditions far away from the urban centers. Because the urban core was no longer a symbol of safety and convenience, high murder and crime rates forced people to migrate to the suburbs, which was also reported by media [49]. In addition, we used Google Earth to check the ground changes in these places. Selected examples are shown in Fig. 11, where SULT-increased pixels covered areas that had built some bungalows. We infer that the increasing human activities in those areas led to the nighttime light increase.

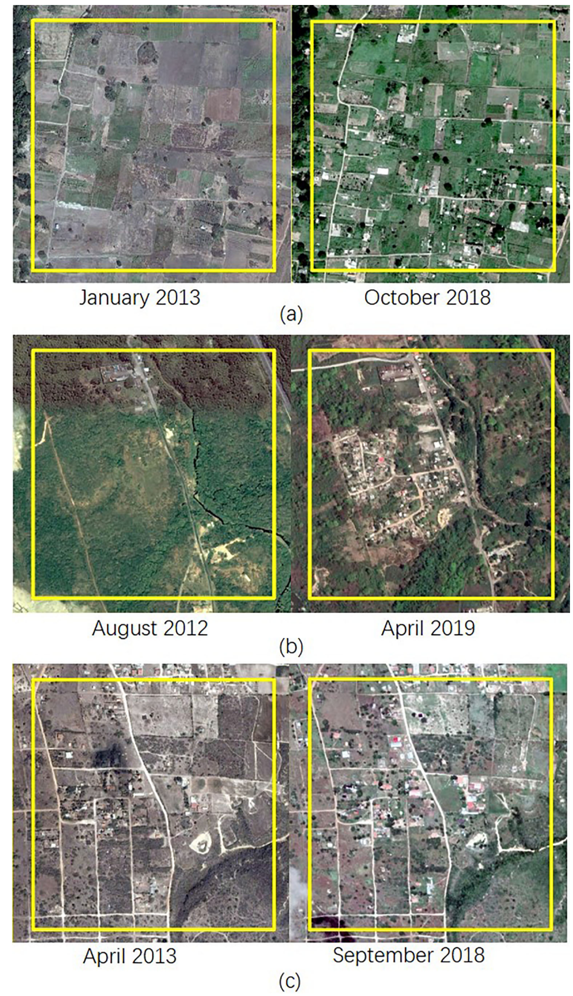

Fig. 11. Google Earth images of SUL-increased suburban areas, a yellow rectangle represents a pixel at a 750 m resolution. (a) Valencia: 10°14′5″N 67°56′59″W. (b) Valencia: 10°25′8″N 68°6′42″W. (c) Barquisimeto: 10°8′34″N 69°16′50″W.

D. Nighttime Light Around the Venezuela–Colombia Border

According to the UNCHR [50], until April 2019, four countries, Colombia, Peru, Chile, and Ecuador, hosted over 65% of the Venezuelan. Among them, Colombia is the top destination for Venezuelan refugees. To analyze the contrast of nighttime light dynamics between Venezuela and Colombia, we also carried out the pixel analysis at a 750-m resolution covering the 60 km buffer area on both sides of the Venezuela–Colombia border (Fig. 12). There were 458 urban lit pixels in the Venezuelan side, which contained 90 pixels and 343 pixels with SULT increased and decreased, respectively. The decrease ratio was 74.89% (Table VI). In contrast, 61.92% of urban lit pixels on the Colombian side had an increase of nighttime light. During the crisis, refugees from Venezuela poured across the Colombian border; this can probably explain the increase in the area of human activity and thereby increased the SULT in those regions [51], [52].

To study the nighttime light dynamics, we divided the 60-km zone on each side of the Venezuela-Colombia border into

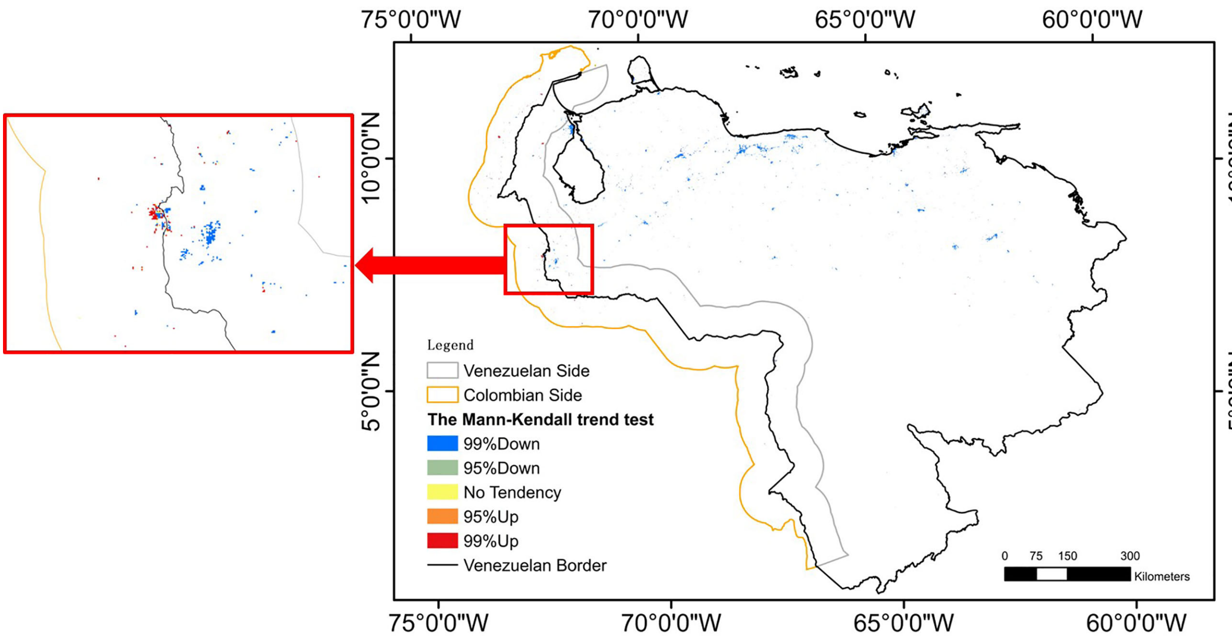

Fig. 12. Pixel analysis showing the spatial pattern of nighttime light change (the full image at high resolution is provided in the supplement).

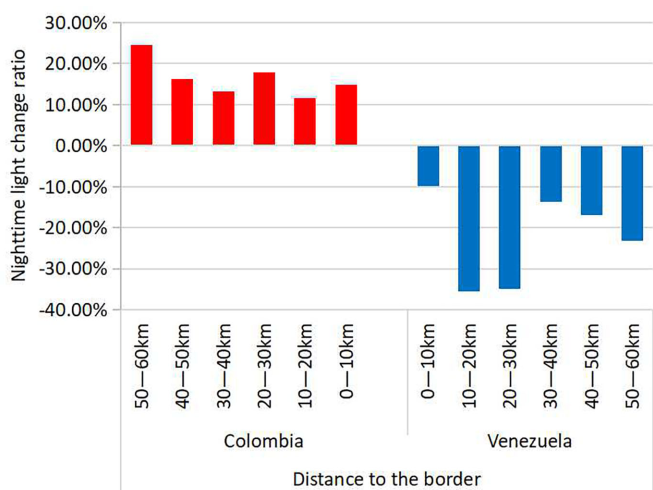

Fig. 13. NLCR of six belts on both sides of the Venezuela–Colombia border.

six belts with a width of 10 km to explore the nighttime light change patterns. In Fig. 13, the six belts in Venezuela had varying degrees of nighttime light loss. Among them, the 10–20 km belt and the 20–30 km belt had the largest decline of nighttime light, and the SULT dropped by over 30%. On the Colombian side, all the six belts had nighttime light increased.

In general, refugees and emigrants had two ways to move to Columbia, by walking and rafting. Due to the high economic vulnerability that prevented them from using conventional transport routes, none of these two emigration methods enabled them to reach the interior of Colombia in a short time. Because of the emigration methods and Colombia's immigration control, Venezuelan refugees and emigrants had a short stay adjacent to the Venezuelan border regions. It may explain why the SULT of regions adjacent to the border dropped relatively slightly.

Taking a detailed look at the border cities, we found 15 cities with urban area larger than 5 km² and within a distance less than 20 km from international borders. The 15 cities include 5 Colombian cities and 10 Venezuelan cities, and some examples are shown in Fig. 14. In those 15 border cities, the SULT of 5 Colombian cities all increased, whereas the SULT of 9 Venezuelan cities decreased except for La Frita, which had no obvious nighttime light change tendency. The nighttime light changed discontinuously crossing the international border from Venezuela into Colombia. Similar results also were found in the Ukraine–Russia border and the South–North Korea border [53].

According to news reports [54]–[56], three Colombian border cities, Cucuta, Arauca, and Maicao, accepted most of Venezuelan refugees. From Table VII, these cities showed a nighttime light increase (Fig. 14). We also measured the distance between these 15 border cities and the border (Table VII). We found that in two Venezuelan cities, Aguas Calientes and El Amparo, the nighttime light did not drop so much compared to other Venezuelan cities; we speculate that refugees did not emigrate immediately, but they stayed in the two cities for a short time, so that the nighttime light did not drop so much. This result is also shown in Fig. 13 where the nighttime light dropped most in the 10–30 km buffer zone of Venezuela and dropped slightly in the 0–10 km buffer zone.

IV. Discussion

We found that the nighttime light dynamic of the border region varies with the distance to the border. We infer that the stay of refugees caused the nighttime light drop not so much in the areas

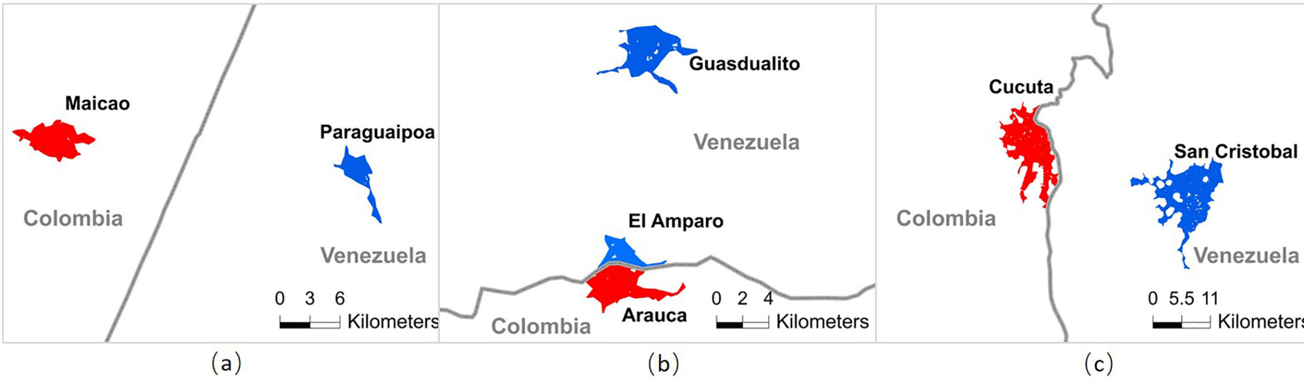

Fig. 14. Samples of Columbia–Venezuela border cities. (a) Maicao-Paraguaipoa. (b) Arauca-El Amparo-Guasdualito. (c) Cucuta-San Cristobal (a red region represents a city with increased SULT; a blue region represents a city with decreased SULT).

adjacent to the international border. However, we need more field investigation data to confirm this possible relationship.

Due to the lack of geographic data on refugee camps, the SULT in the refugee camps and their surrounding areas is hard to calculate. Therefore, the SULT analysis in the buffer zone around the Venezuelan–Colombian border includes refugee and nonrefugee activities. Considering the above reasons, the direct impact of refugees on the nighttime light of the Colombian border areas needs further analysis.

However, Venezuela has experienced many events of electricity power failures, which had significant impacts on nighttime lights. If we can obtain the power blackout and electricity-related data, we can further analyze the impact of the power blackout on the daily nighttime light. Although we used HP-filter to remove short time fluctuation, the atmospheric environment, plant phenology as well as satellite viewing angles [57] will also affect the changes in nighttime light, we need to study the effect of these two factors on the nighttime light in future.

As the number of Venezuelan refugees is still rising, many NGOs are carrying out aid projects [58], and more refugee camps will be established in the future. Based on the availability of refugee data, nighttime light in the refugee camps and its surrounding areas will be further investigated and would be potentially useful for the international community, such as helping UNHCR to detect the regions where residents are affected by the crisis, and to decide where to build refugee camps, and to provide emergency aid with geographic information.

V. Conclusion

Using VIIRS nighttime light imagery, we found that Venezuela had lost 30.37% of urban light during April 2012 to December 2018. Among the 24 administrative areas, 19 administrative areas lost their urban nighttime light by more than 30%. The regression results show that oil production had a positive correlation with nighttime lights, but the exchange rate and number of asylum seekers had negative correlations with nighttime lights. These findings may extend nighttime light applications, while nighttime light data were used to estimate GDP, electricity consumption, and regional development in previous studies [59]. In addition, we found that the most of oil refineries' nighttime light had decreased by varying degrees, but a small number of PDVSA-associated refineries' nighttime light had increased. The pixel analysis showed that some suburban areas experienced an increase in nighttime lights, indicating that residents moved from the urban center to the suburbs. Also, we revealed the nighttime light change pattern of the Venezuela–Colombia international border, which was possibly impacted by the movement of the refugees. In summary, the spatiotemporal information from nighttime light images would be a good data source to evaluate Venezuela's crisis.

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